

EduRAG

PROBLEM STATEMENT

Long lecture videos are difficult to search precisely. Students remember fragments like a diagram, code snippet, or spoken phrase, but not the exact timestamp.

Need

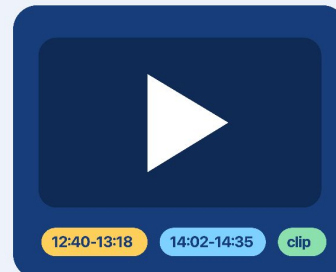
Find the exact explanation in seconds, with evidence and timestamp, instead of searching through the whole lecture.

- Transcript-only search misses visual and time-dependent explanations. Revision becomes slow because learners must rewatch large parts of the lecture video.

WORKING PROTOTYPE

Where is yield point explained?

How the prototype works



System response

Yield point is discussed when the stress-strain curve moves from elastic to plastic behavior. The learner gets grounded clips and can verify the answer directly inside the lecture.

Grounded answer Timestamps Context

Prototype capabilities

- Natural-language question answering
- Timestamp-linked clip retrieval
- Follow-up queries and session continuity

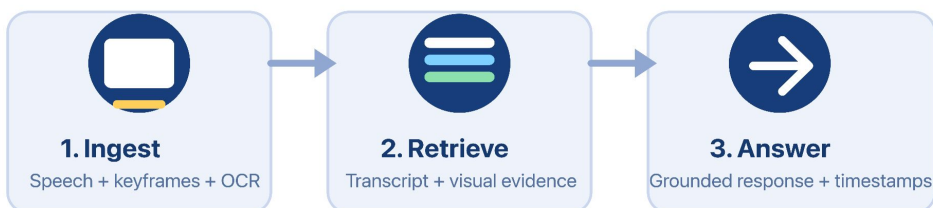
What the user experiences

Ask a lecture question, receive a grounded answer, and jump directly to the relevant explanation instead of manually searching the whole video.

ABSTRACT

EduRAG turns lecture videos into searchable learning material

A multimodal RAG system for question answering with timestamps



What the user gets

Natural-language QA Evidence grounding Jump-to timestamps Follow-ups

METHODOLOGY & TECHNICAL STACK

Pipeline



Core idea

Use both spoken content and visual evidence, then generate answers only from retrieved lecture context.

Technical stack

Python FastAPI Faster-Whisper ChromaDB
NetworkX OCR Embeddings Graph React
Ollama Gemini Sentence Transformers

RESULTS

Evaluation results

15 questions, 3 lecture videos, DeepEval with local judge

Dataset slice

3 domains
15 QA pairs

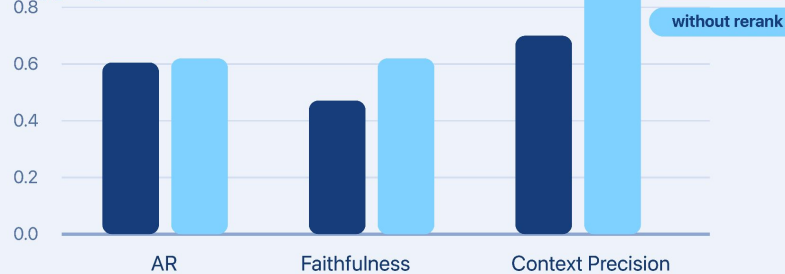
Metrics

AR, Faithfulness,
Context Precision

Main finding

Without rerank performed better on this slice

Aggregate comparison



Takeaway: the pipeline works, but reranking still needs calibration for broader educational queries.

CONCLUSION



Better access

Students can ask natural questions and jump to the explanation quickly.



Grounded answers

Responses are tied to retrieved evidence and timestamped video.



Next step

Improve reranking, expand benchmark, and add stronger models.

Educational videos should not behave like static archives. EduRAG turns them into interactive learning tools.



TEAM

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